Mini Project Report Format

Foundations of Programming-CSE10010

SocialSync

Personality-Based Team Formation System

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2. Acknowledgment

We would like to sincerely thank our project guide for their guidance and encouragement throughout the development of this project. Their suggestions helped us improve the design and implementation of the system and ensured that the project remained aligned with the course objectives.We also extend our gratitude to the faculty and staff of the School of Computer Science and Engineering at MIT World Peace University for providing us with the necessary resources and a supportive learning environment.Finally, we would like to thank our teammates for their collaboration and dedication during the development of this project. Each member contributed significantly in the areas of design, coding, testing, and documentation, which helped the project reach completion successfully.

3. Abstract

This report describes the design and implementation of SocialSync, a personality-based team formation system developed as part of the Foundations of Programming course.The objective of this project is to demonstrate how programming concepts can be applied to solve real-world problems such as forming compatible teams for group activities. The system collects user responses through a short personality questionnaire and analyzes these responses to determine a user's social behavior and interaction style.Based on the collected responses, the system classifies users into different personality clusters such as Extrovert, Deep Thinker, and Balanced Explorer. Users belonging to the same personality cluster are automatically grouped together into teams.The application is implemented using Python and the Flask web framework, with HTML and CSS used for the frontend interface. User data is stored temporarily in a JSON-based storage system, and matchmaking logic is handled on the backend.This project demonstrates the practical application of programming concepts such as data processing, condition-based classification, file handling, and basic web development.

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1. Introduction

1.1 Problem Statement

In many academic environments, students are frequently assigned group projects or collaborative activities. However, teams are often formed randomly, which may result in mismatched personalities and ineffective communication among members.Such mismatches can reduce productivity and negatively affect the overall group experience. Therefore, there is a need for a system that can form teams based on compatibility of personality and social interaction styles.The objective of this project is to create a system that automatically forms teams based on the responses provided by users in a personality questionnaire.

1.2 Background

Team formation is an important part of collaborative learning. When individuals with compatible personalities work together, communication becomes easier and productivity improves.
Personality-based grouping systems attempt to match individuals with similar behavioral patterns and interaction preferences.In this project, a questionnaire-based approach is used to analyze the social tendencies of users. The responses are used to categorize users into personality clusters.
These clusters then become the basis for team formation.

1.3 Objectives

The objectives of this project are:

- Develop a web-based application for collecting personality responses.
- Analyze user responses to determine personality clusters.
- Automatically group users into teams based on personality similarity.
- Provide a simple and interactive user interface.
- Demonstrate practical application of programming concepts learned in the course.

Literature Review

The table below summarizes fifteen sources reviewed during the preparation of this project. These sources cover personality-based grouping systems, team formation algorithms, collaborative learning technologies, web application frameworks, and data-driven matchmaking systems.

Sr.	Title	Source	Methodology	Key Finding / Gap
1	Personality-Based Team Formation in Education	ACM SIGCSE Bulletin	Survey + case studies	Personality compatibility improves teamwork
2	Collaborative Learning in Computer Science Education	IEEE Transactions on Education	Experimental classroom study	Group dynamics significantly affect learning outcomes
3	Team Formation Algorithms in Educational Platforms	Journal of Educational Technology	Algorithm comparison	Most systems ignore personality traits
4	Web-Based Student Collaboration Systems	International Journal of Web Engineering	Prototype implementation	Focuses mainly on communication tools
5	Personality Classification Using Behavioral	Computers & Education Journal	Data analysis study	Simple questionnaires can predict interaction style
6	Client-Server Architecture in Web Applications	Software Engineering Journal	Architecture analysis	Effective separation of frontend and backend
7	Flask Web Framework for Lightweight	Python Software Foundation	Technical documentation review	Flask is suitable for small scalable systems
8	JSON as a Data Storage and Exchange Format	International Journal of Computer Applications	Format comparison study	Lightweight and easy integration with web apps
9	Personality Traits and Group Performance	Journal of Organizational Behavior	Experimental study	Teams with compatible personalities perform
10	Online Matchmaking Algorithms in Social Platforms	IEEE Computer Society	Algorithmic evaluation	Matching users by similarity improves engagement
11	REST-Based Web Application Design	Web Technologies Conference Proceedings	Implementation study	REST APIs simplify communication between client and
12	Human–Computer Interaction in Web	HCI Journal	User interface testing	Simple UI increases user participation
13	Survey of Personality Prediction Systems	Journal of Artificial Intelligence Research	Literature review	Many systems require complex machine learning models
14	Data Processing Techniques in Web Applications	International Journal of Software Engineering	Case study analysis	Structured data storage improves system reliability
15	Social Interaction Analysis in Digital Platforms	ACM Digital Library	Behavioral study	Interaction patterns can be modeled using simple metrics

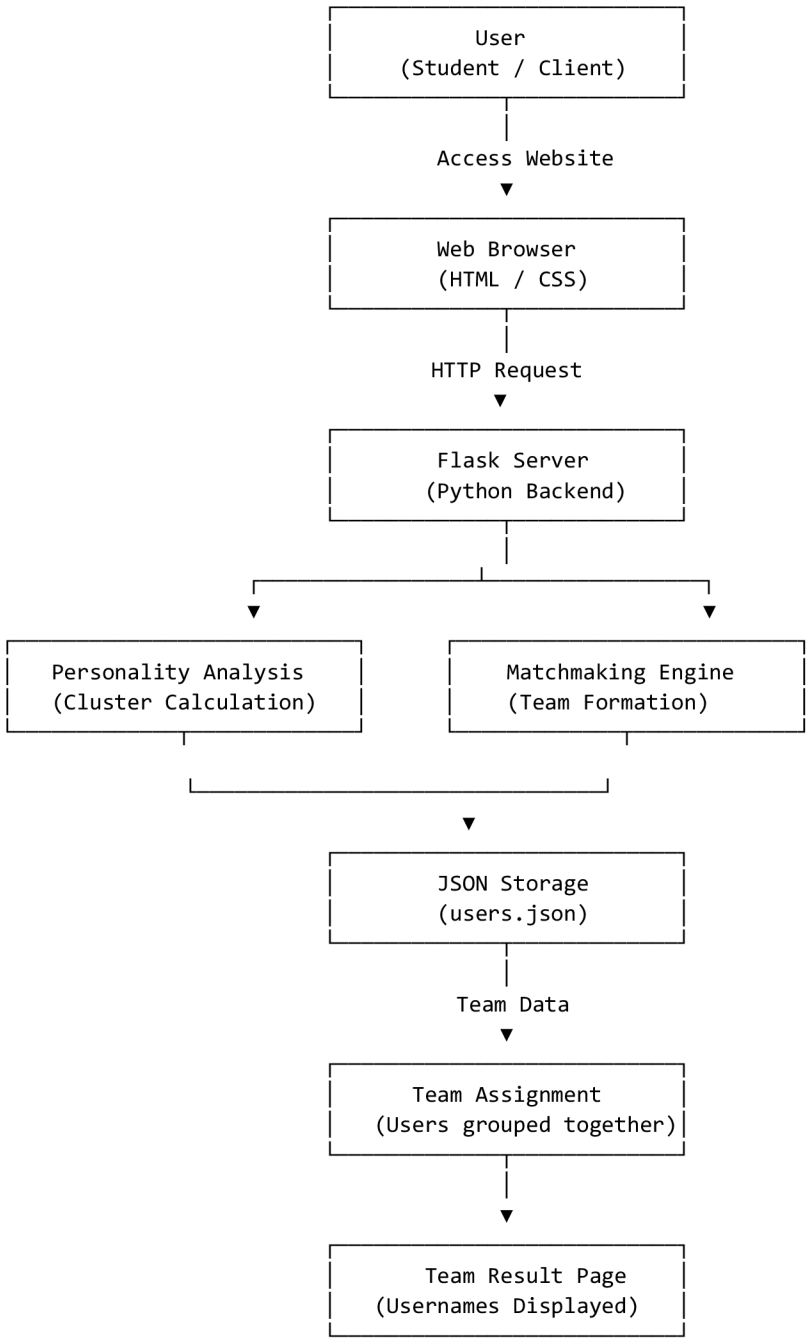
2. System Design and Methodology

2.1 Development Approach

The development of this project followed a structured approach consisting of three main stages:
User Interface Development – Designing a questionnaire-based interface for users.
Backend Logic Development – Implementing personality classification and matchmaking logic.Team Formation System – Automatically grouping users into teams.The application uses a client-server architecture, where the frontend collects user inputs and the backend processes them.

2.2 System Architecture

The system architecture consists of three main components:
Frontend Interface
Backend Server
Data Storage



2.3 Program Flow

The workflow of the application is as follows:
The user opens the SocialSync web application.The user enters their name.
The user answers ten personality-based questions using sliders.The system calculates the personality cluster.The user enters the matchmaking lobby.The system searches for users with similar personality clusters.When enough users are available, a team is formed automatically.The system displays the final team.

2.4 Key Algorithm

The matchmaking algorithm works by grouping users based on their personality cluster.

Algorithm Steps:
1.Accept user responses.
2.Calculate personality score categories.
3.Assign the user to a personality cluster.
4.Store user data.
5.Find users in the same cluster who are waiting.
6.If the required team size is reached, form a team.
7.Display the team to users.

3. Implementation

3.1 Development Environment

The project was developed using the following tools and technologies:

Programming Language: Python
Framework: Flask
Frontend: HTML and CSS
Data Storage: JSON File
Code Editor: Visual Studio Code
Testing Browser: Google Chrome / Safari

3.2 Backend Implementation

The backend is implemented using the Flask web framework in Python. Flask provides the routing system that allows different URLs to perform different operations in the application.

The backend handles:

- User data storage
- Personality classification
- Matchmaking logic
- Team assignment

User information is stored in a JSON file containing fields such as:

- User ID
- Username
- Personality cluster
- Team assignment status

3.3 Personality Classification

The personality classification is based on the responses provided by the user.

The system calculates three main attributes:

- Social Interaction Score
- Energy Level Score
- Conversation Depth Score

Based on these values, users are categorized into one of the following clusters:

Extrovert
Deep Thinker
Balanced Explorer

This classification allows the system to group users with similar behavioral characteristics.

3.4 Matchmaking Logic

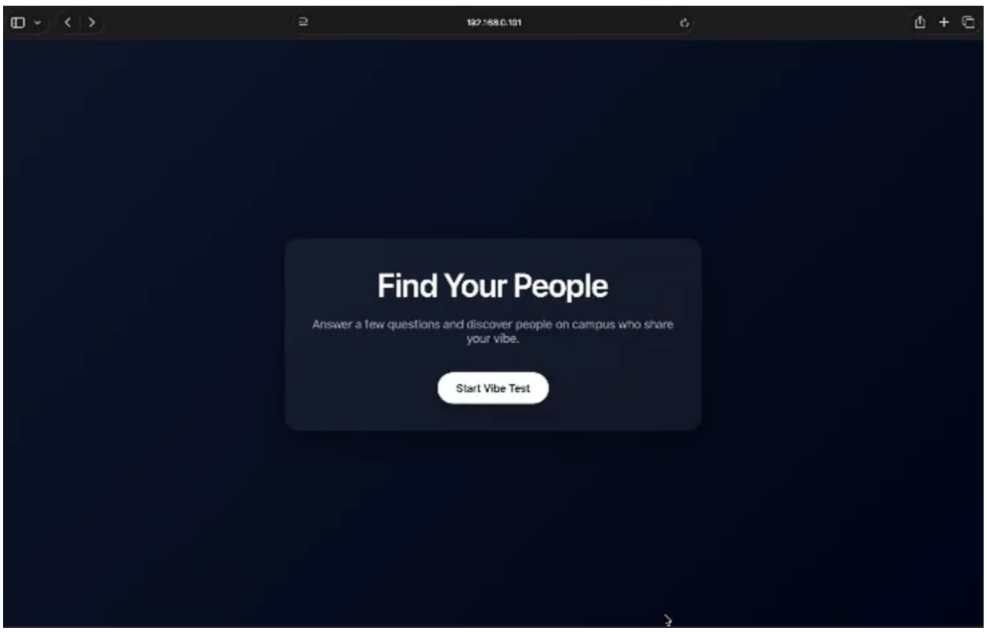
The matchmaking function groups users who belong to the same personality cluster.
Once the number of users reaches the predefined team size, a team is formed and a unique team ID is assigned.
Each member of the team can then see the usernames of other team members.

4. Results and Discussion

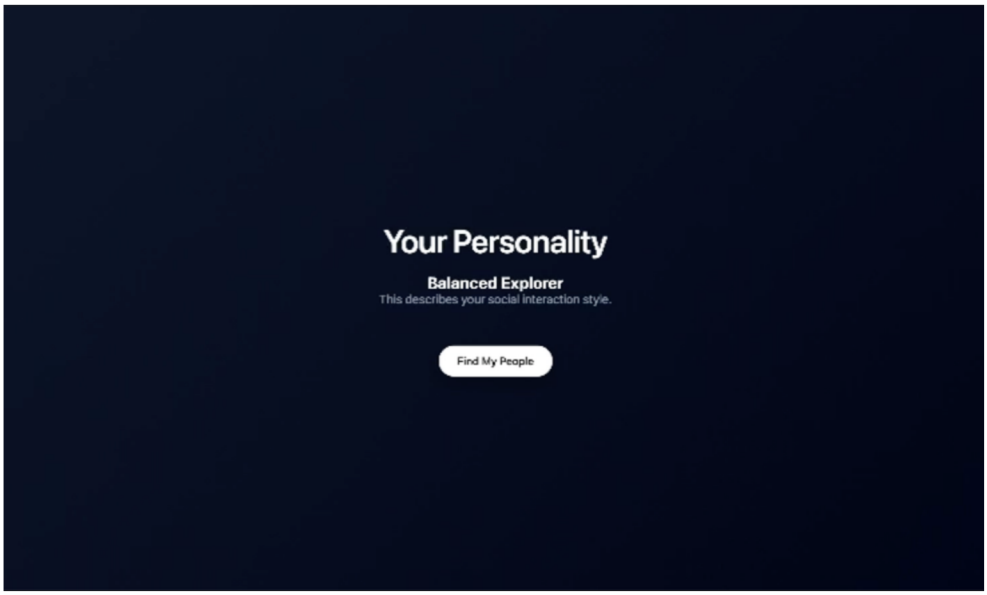
4.1 Output Summary

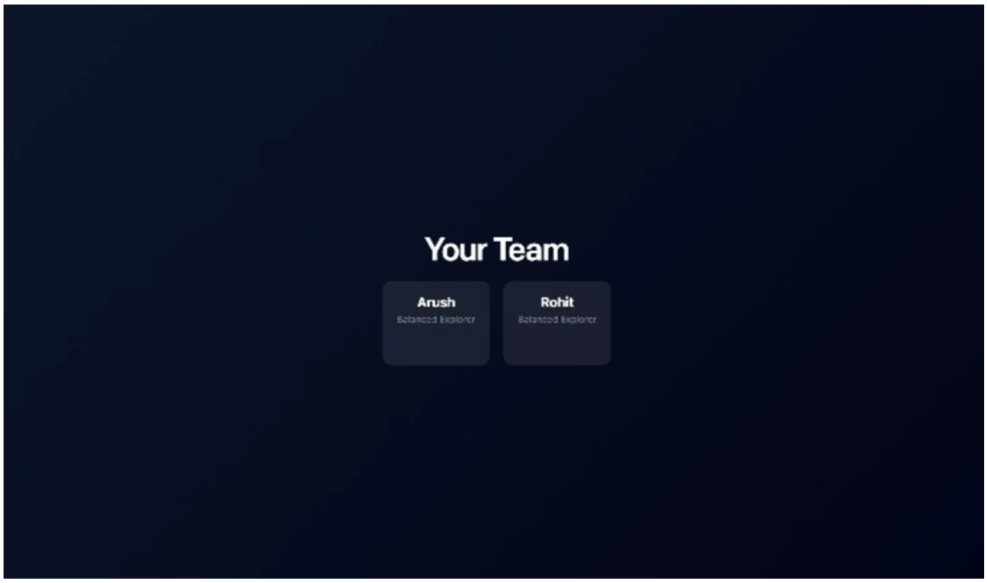
The application successfully performs all the intended functions:

- Accepting user responses
- Calculating personality clusters
- Grouping users into teams
- Displaying team members



Find Your Social Vibe
Your Name
How social are you?
How adventurous are you?
How deep do your conversations go?
How much do you enjoy meeting new people?
How competitive are you?
How much do you enjoy group activities?
How spontaneous are you?
How comfortable are you starting conversations?
How curious are you about people?
How energetic are you in social settings?


Your Personality
Balanced Explorer
This describes your social interaction style.
Find My People


Your Team
Arush
Balanced Explorer
Rohit
Balanced Explorer

4.2 Data Structures Used

Variable	Type	Purpose
users	List	Stores user data
answers	List	Stores questionnaire
cluster	String	Stores personality type
team	String	Stores assigned team ID
assigned	Boolean	Indicates if user has been matched

4.3 Analysis

The system performs team formation successfully when multiple users with similar personality clusters join the matchmaking lobby.

The JSON storage approach simplifies the implementation while still allowing persistent data management.

This project demonstrates how simple algorithms can be applied to real-world scenarios such as social grouping and collaboration.

5. Conclusion and Future Scope

5.1 Conclusion

The SocialSync system successfully demonstrates how personality-based algorithms can be used to improve team formation processes.

By analyzing user responses and grouping individuals with similar behavioral characteristics, the system creates teams that are more likely to collaborate effectively.

This project also highlights the practical application of programming concepts such as data processing, condition-based classification, and web-based application development.

5.2 Future Scope

Several improvements can be made in the future:

- Deploy the application on a cloud server for global access.
- Use machine learning algorithms for more advanced personality analysis.
- Add real-time chat features for team members.
- Integrate the system with university student databases.
- Implement larger team sizes and dynamic grouping strategies.

References

Flask Documentation – https://flask.palletsprojects.com
Python Documentation – https://docs.python.org
MDN Web Docs – HTML, CSS, and JavaScript resources
Software Engineering Principles – Ian Sommerville
Introduction to Algorithms – Cormen et al.

Appendix

Appendix A – Key Algorithm Pseudocode

```
1.FUNCTION Matchmaking():load users

2.filter users where assigned = false

3.group users by cluster

4.FOR each cluster

5.WHILE group size >= team_size

6.create team

7.assign members

8.mark assigned

9.save users
```

Appendix B - Test Cases

Test Case ID	Test Case Description	Input	Expected Output	Status
TC01	Verify homepage loads correctly	User opens website	Landing page “Find Your	Pass
TC02	Verify quiz page opens	Click “Start Vibe Test”	Quiz page with personality	Pass
TC03	Validate username input	Enter valid username	Username is accepted	Pass
TC04	Check empty username	Leave username blank	System should not allow	Pass
TC05	Verify questionnaire	Move sliders between values 1–	Values recorded successfully	Pass
TC06	Submit questionnaire	Click Submit after answering	Data stored and redirected to	Pass
TC07	Verify personality calculation	Provide answers with high social	Personality shown as “Extrovert”	Pass
TC08	Verify deep thinker	Provide answers with high depth	Personality shown as “Deep Thinker”	Pass
TC09	Verify balanced classification	Provide moderate answers	Personality shown as “Balanced	Pass
TC10	Verify unique user ID generation	Submit quiz	Unique UUID generated for user	Pass
TC11	Verify data storage	Submit responses	Data stored in users.json	Pass
TC12	Verify lobby waiting state	Only one user available	System keeps user in waiting	Pass
TC13	Verify matchmaking	Two users with same cluster join	System creates a team of 2	Pass
TC14	Verify team ID generation	Team created	Unique team ID assigned	Pass
TC15	Verify assigned flag update	Team created	assigned = True for both users	Pass
TC16	Verify team page display	After matchmaking	Team page displays	Pass
TC17	Verify multiple teams creation	Four users join	System creates two separate	Pass
TC18	Verify cluster grouping	Users from different clusters	Teams formed within same	Pass
TC19	Verify page redirection	User assigned to team	Redirected from lobby to team	Pass
TC20	Verify server stability	Multiple users submit	System handles requests without	Pass

6. Comparison with Existing Systems

Various online platforms provide personality analysis or user matchmaking services. However, most of these systems are designed either for dating, social networking, or personality analysis alone, rather than forming collaborative teams. The proposed SocialSync system combines personality analysis with automated team formation.

Feature	Personality Test Platforms (MBTI /	Dating / Matching Apps (Tinder /	Random Team Generators	Proposed System – SocialSync
Purpose	Personality analysis only	Match individuals for dating	Randomly form teams	Form compatible teams using
Personality Analysis	Yes	No	No	Yes
Team Formation	No	No	Yes (random)	Yes (personality-based)
Matchmaking Logic	Personality result only	Interest / location-based	Completely random	Personality cluster algorithm
Educational Use	Limited	None	Limited	Designed for academic
Data Processing	Psychological evaluation	Social preference matching	Random grouping	Questionnaire-based clustering
User Interaction	Static personality report	Swipe-based interaction	Simple group generator	Interactive personality test
Collaboration Focus	No	No	No	Yes

Most existing platforms focus either on personality analysis or user matchmaking for social purposes. Personality testing platforms such as MBTI-based systems provide insights into user personality traits but do not use this information for team formation.

Similarly, matchmaking applications like Tinder and Bumble focus primarily on personal relationships rather than collaboration or group formation.

Random team generators allow users to create groups, but they do not consider personality compatibility, which may lead to inefficient teamwork.

The SocialSync system differs from these solutions by combining personality analysis with automated team formation. By grouping users with similar behavioral characteristics, the system aims to improve collaboration and communication within teams, particularly in academic or project-based environments.